A continuum charge-transport index for the lowest Landau level

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Abstract

We give a continuum version of the charge-transport argument of Bachmann–Bols–De Roeck–Fraas, arXiv:1810.07351. The arrangement of the argument is retained: spatial and charge data, an index theorem, verification from a spectral gap, examples, and the boundary-factorization proof. A finite-rank formulation is proved as well, using a determinant instead of the phase of a single state. The lowest Landau level is included in the ordinary continuum fermion space, and particles outside it are assigned an auxiliary energy proportional to the volume. Rapid physical-space locality of the interaction and Gaussian projection kernels supply locality, while physical regional particle numbers retain integer spectra. The construction uses no localized orthonormal basis and no discrete tight frame. The low-energy sector is a band of d states with width at most w, separated from higher states by a uniform gap $\gamma > w$. Exact degeneracy is not assumed; exponentially small splitting is included. The interaction has a uniformly bounded, neutral LLL density with local ambient approximants whose errors decay faster than every power of the radius. No bound on its polynomial degree or coefficients is imposed. An extensive auxiliary penalty preserves the band even at saturation; periodicity of the LLL number evolution prevents this penalty from spoiling the macroscopic propagation estimates. Neither positivity, zero ground energy, nor a particle-removal estimate is needed. Magnetic translation gives the constraint $dn_\sigma/N_\phi \in \mathbb{Z} + O(L^{-\infty})$, where d is the rank of this band in the selected spin-number sector. At balanced rational filling $\nu = p/q$ in lowest terms, d is a multiple of $2q/\gcd(p, 2)$: this is $2q$ for odd p, and q for even p. The same divisibility holds for the full low-energy band at fixed total particle number when the Hamiltonian has spin-exchange symmetry. With full SU(2) symmetry, such a band consists entirely of spin singlets for all sufficiently large tori. Appendices derive the required kernel, propagation, and spectral-filter estimates. No gap at nonzero boundary twist or local accessibility of the first excited state is assumed.

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1 Introduction

The index in [1] measures charge transported through one of two distant boundaries by a charge-conserving unitary. Its proof uses a dressed regional charge, separation of the two boundaries, and the periodicity of an integer-valued charge. Those are the ingredients we preserve. For a low-energy sector of rank $d > 1$, the appropriate quantity is an unnormalized trace. We prove this extension explicitly; the determinant formulation is motivated by [2], but no theorem from either reference is used without proof here. As in the finite-rank formulation of [2], this sector may contain several distinct eigenvalues. The gap used in the proof is above the entire sector, rather than between its individual levels.

There are two separate issues in transferring this argument to a Landau level. First, projecting a physical region onto the LLL does not give a projection. Its second quantization therefore need not be an integer-valued charge. Second, a spatially localized phase rotation does not preserve the LLL. We retain the full spatial Hilbert space to resolve both issues, and penalize its additional modes to preserve the desired low-energy spectrum.

We work with self-adjoint, neutral LLL densities that are uniformly bounded on the full Fock space and admit physical local approximants with errors smaller than every inverse power of their radius. Uniformly bounded field polynomials with Gaussian-local test vectors are examples, but neither a degree bound nor a bound on their polynomial coefficients is required. A penalty proportional to the volume preserves the selected band using only the extensive norm bound on the Hamiltonian, including at saturated fillings. Its large coefficient does not force a large propagation speed in the estimates we need: the penalty dynamics factors through the integer-valued LLL number operator and can be reduced to a bounded time interval.

Two further ingredients handle the weaker locality hypothesis. Truncating interaction range and evolution time at $L^{1/3}$ gives superpolynomial accuracy across macroscopic buffers. Finite LLL rank bounds the regional-charge Fourier bandwidth, allowing bounded current approximants without

commuting an arbitrary local operator with an unbounded charge. The resulting currents have norm $O(L^3)$, which is sufficient for the index argument.

Only magnetic translations at the original boundary conditions are needed. The auxiliary parameter in the index proof parametrizes exponentials of dressed charges; it is not a family of Hamiltonians whose spectral gap has to be established.

We write d for the low-energy band rank, reserving p, q for the coprime numerator and denominator of the total filling. The general spin-resolved constraint is derived first. We then specialize to balanced populations, discuss the numerator and denominator parity, and prove a fixed-total-number version using spin-exchange symmetry. The interaction class covers balanced fillings $0 < \nu \leq 2$, including saturation without an additional energy-comparison assumption. The rank d is fixed and the gap above the band is uniform in the selected number sector. The exact multiplicity of the lowest eigenvalue may be smaller than d.

2 The index theorem and its LLL setting

2.1 Spatial setup, fermions, and charges

Let

$$\mathbb{T}_L = \mathbb{R}^2/(L_x\mathbb{Z} \oplus L_y\mathbb{Z}), \qquad L = \min(L_x, L_y), \qquad c_* \leq L_x/L_y \leq C_*,$$

where $L \to \infty$. Work in units with magnetic length one, with

$$M = N_\phi = \frac{L_x L_y}{2\pi} \in \mathbb{N}.$$

The one-particle space $\mathfrak{h}$ is the space of square-integrable sections of the magnetic line bundle. In the gauge $A = (0, x)$ its sections obey

$$f(x + L_x, y) = e^{\mathrm{i}L_x y} f(x, y), \qquad f(x, y + L_y) = f(x, y). \tag{1}$$

Let $\mathcal{F} = \mathcal{F}_{\mathrm{a}}(\mathfrak{h} \otimes \mathbb{C}^2)$, the fermionic Fock space for spin-1/2 electrons. The annihilation operators are antilinear in their argument and satisfy

$$\{a_\sigma(f), a^\dagger_{\sigma'}(g)\} = \delta_{\sigma\sigma'} \langle f, g\rangle , \qquad \|a_\sigma(f)\| = \|f\|_2 .$$

Set $N_\sigma = \mathrm{d}\Gamma_\sigma(1)$, and fix the sector

$$\mathcal{H}_{\boldsymbol{n}} = \{\psi : N_\sigma\psi = n_\sigma\psi,\ \sigma = \uparrow, \downarrow\}, \qquad n_\uparrow + n_\downarrow \leq CL^2.$$

A bounded operator is *local in* X if it acts on $\mathcal{F}_{\mathrm{a}}(L^2(X) \otimes \mathbb{C}^2)$, tensored with the identity on the complement, with the usual fermionic grading. We use only even operators, so operators in disjoint regions commute. An operator is *neutral* if it commutes with both particle numbers. Locality is always defined in the ambient Fock space, before restricting to $\mathcal{H}_{\boldsymbol{n}}$.

For a measurable region X, define the physical spin-up charge

$$Q_X = \mathrm{d}\Gamma_\uparrow(1_X).$$

Although Q_X is unbounded on $\mathcal{F}$, its exponential is a bounded local unitary, and

$$e^{2\pi\mathrm{i}Q_X} = 1, \qquad 0 \leq Q_X|_{\mathcal{H}_{\boldsymbol{n}}} \leq n_\uparrow. \tag{2}$$

All charges commute. Conjugation by $e^{\mathrm{i}sQ_X}$ multiplies the spin-up test function by a phase of modulus one and preserves its spatial support.

An estimate $A \simeq B$ means $\|(A-B)|_{\mathcal{H}_{\boldsymbol{n}}}\| \le C_k L^{-k}$ for every k. When a bound is in ambient Fock-space norm we say so explicitly. All constants may depend on the fixed density bound, aspect-ratio bounds, interaction, band rank, and spectral bounds (including the separation margin $\gamma - w$), but not on L. Polynomial factors in L, and square roots, preserve an $O(L^{-\infty})$ error.

Take

$$\Gamma = [0, L_x/2) \times S^1_y, \qquad Q = Q_\Gamma.$$

Its boundary components are denoted ∂_- and ∂_+, at $x = 0$ and $x = L_x/2$, respectively. Let $S_\pm$ be the strips of transverse radius $L_x/8$ about these boundaries, and let $B_\pm$ be the narrower strips of radius $L_x/16$. Define

$$Q_\pm = Q_{\Gamma \cap S_\pm}, \qquad Q_m = Q - Q_- - Q_+.$$

The two $S_\pm$ are separated by a distance proportional to L.

2.2 Hypotheses and statement of the index theorem

Let $P = P_L$ be an orthogonal projection of fixed finite rank d in $\mathcal{H}_{\boldsymbol{n}}$. The following hypotheses isolate the algebraic content of the index proof. The conditions on U and its boundary charge transport correspond to Assumption 2 ("Charge and locality preserving U") of [2]. Here we state the boundary-locality properties needed in the proof explicitly, and also include the dressing and clustering hypotheses that are established from the spectral gap below.

Assumption 2.1 (Hypotheses of the index theorem)**.**

1. U is an ambient unitary preserving both particle numbers, and $[U, P] \simeq 0$.

2. There are neutral boundary transports $T_\pm$, localized in $B_\pm$, with polynomial sector norms, such that
$$U^\dagger Q U - Q \simeq T_- + T_+.$$
The operators $Q_\pm + T_\pm$ have self-adjoint local realizations and integer spectra:
$$e^{2\pi \mathrm{i}(Q_\pm + T_\pm)} = 1. \tag{3}$$
Here and below unbounded local charges are used through their bounded sector restrictions or their ambient unitary exponentials.

3. There are bounded, self-adjoint, neutral $K_\pm$, localized in $B_\pm$, with polynomial ambient norms, such that
$$[Q - K_- - K_+, P] \simeq 0.$$
There are also self-adjoint, neutral $K^U_\pm$, localized in $B_\pm$, with polynomial ambient norms, satisfying
$$K^U_\pm \simeq U^\dagger K_\pm U.$$

4. For bounded, neutral ambient operators A, B supported in S_-, S_+, respectively, there is a uniform estimate
$$\|PA(1-P)BP\| \le \|A\|\,\|B\|\,O(L^{-\infty}). \tag{4}$$

The normalization (3) is an explicit hypothesis. We do not infer it from a tensor-factor argument inside a fixed-particle-number sector. It will hold exactly for magnetic translation. Spatial support in this assumption means ambient support, not support of a compression to the LLL.

Theorem 2.2 (Continuum transport index)**.** *Under Assumption 2.1,*

$$\operatorname{dist}\big(\operatorname{Tr}(PT_-),\mathbb{Z}\big) = O(L^{-\infty}). \tag{5}$$

Consequently the normalized transport $\operatorname{Ind}_P(U) = d^{-1}\operatorname{Tr}(PT_-)$ *lies within* $O(L^{-\infty})$ *of* $d^{-1}\mathbb{Z}$.

The proof in Section 4 follows closely the structure of [1, Section 4]. For $d = 1$, (4) is ordinary connected clustering and the calculation concerns a scalar phase, as in that paper. For general d, approximate eigenvector statements become approximate invariance of $\operatorname{Ran} P$, and the scalar phase is replaced by the determinant of the boundary unitary compressed to $\operatorname{Ran} P$. This finite-rank argument is already used in the proof of [2, Theorem 3.2, Section 4]; the determinant construction here is an adaptation of that proof. We compute its derivative directly by Jacobi's formula. At 2π, the compressed unitary is approximately a group commutator, whose determinant is one, yielding (5).

The continuum adaptation keeps spatial locality on the full Fock space while estimating unbounded charges in a fixed-number sector. The LLL-specific work is the verification of the hypotheses below, rather than a different mechanism for integrality. We never apply the rank-one theorem separately to vectors in a higher-rank subspace, and no equality of energies within $\operatorname{Ran} P$ is required by this abstract theorem or its proof.

2.3 The LLL and an exact enlargement

Put

$$D_x = \partial_x, \quad D_y = \partial_y - \mathrm{i}x, \quad D = \tfrac{1}{2}(D_x - \mathrm{i}D_y).$$

Let Π be the orthogonal projection onto $\mathfrak{h}_0 = \ker(D_x + \mathrm{i}D_y)$, of dimension M. For $R \in \mathbb{T}_L$, its evaluation and derivative-evaluation vectors are determined by

$$\langle e_R, f\rangle = f(R), \qquad \langle g_R, f\rangle = (Df)(R), \qquad f \in \mathfrak{h}_0.$$

They belong to $\mathfrak{h}_0$, even though Df need not. Appendix A proves, uniformly in the torus,

$$\|e_R\| + \|g_R\| \le C, \qquad \left\|1_{\{d(\cdot,R)>r\}} e_R\right\| + \left\|1_{\{d(\cdot,R)>r\}} g_R\right\| \le Ce^{-cr^2}, \qquad \int_{\mathbb{T}_L} |e_R\rangle\langle e_R|\,\mathrm{d}R = \Pi. \tag{6}$$

Definition 2.3 (Bounded rapidly local LLL interaction)**.** Let

$$H = H_{\mathrm{LLL}} = \int_{\mathbb{T}_L} h_R\,\mathrm{d}R.$$

Each $h_R = h_R^\dagger$ is a neutral operator on the LLL Fock factor, canonically extended by the identity on its orthogonal-complement Fock factor, and $\|h_R\| \le C$ uniformly in R, L. The density is measurable. For every $r \ge 1$ there are measurable, self-adjoint, neutral ambient operators $h_R^{(r)}$, supported in the physical ball $B_r(R)$, such that for every k

$$\left\|h_R - h_R^{(r)}\right\| \le C_k(1+r)^{-k}. \tag{7}$$

All constants are independent of R, L, r. At radii covering the torus, take $h_R^{(r)} = h_R$. Weak operator measurability of the approximants suffices; their integrals below may be taken in the weak operator sense. Uniform norm bounds control those integrals.

The approximants need not preserve the LLL. Their spatial support and the error in (7) are measured on the full physical Fock space, before selecting a number sector. Averaging an approximant over both global gauge rotations preserves its support and makes it neutral without increasing its error. For the finite-dimensional LLL algebra, measurability of h_R also gives a norm integral. No positivity or uniform polynomial-degree assumption is imposed.

For example, a self-adjoint neutral polynomial in fields with test vectors e_R, g_R, or a fixed finite family of uniformly bounded Gaussian-local LLL vectors, satisfies this definition when its degree, number of monomials, and coefficients are uniformly bounded. Truncating the test vectors to $B_r(R)$ and telescoping each product gives a Gaussian version of (7) by the CAR. This is a sufficient subclass, not a restriction on the general density. The locality assumption concerns physical space; no equivalence with locality in an orbital or slot embedding is asserted.

The class also includes projected two-body potentials with uniformly integrable, rapidly decaying tails. For example, let $\mathcal{V}_L$ be a real even potential on the torus with

$$\sup_L \int_{\mathbb{T}_L} (1 + d(0,S))^k |\mathcal{V}_L(S)| \, \mathrm{d}S < \infty \qquad \text{for every } k \geq 0.$$

Periodizing a real even potential on $\mathbb{R}^2$ with all weighted L^1 moments gives such a family. Define

$$h_R = \frac{1}{2} \sum_{\sigma,\sigma'} \int_{\mathbb{T}_L} \mathcal{V}_L(R-S) a^\dagger_\sigma(e_R) a^\dagger_{\sigma'}(e_S) a_{\sigma'}(e_S) a_\sigma(e_R) \, \mathrm{d}S.$$

The CAR give $\|h_R\| \leq C \, \|\mathcal{V}_L\|_1$. To obtain an approximant supported in $B_r(R)$, retain only centers with $d(R,S) \leq r/2$ and restrict both test vectors to $B_r(R)$; once that ball covers the torus, take the exact density instead. Each retained monomial is of the form $B^\dagger B$, so real coefficients preserve self-adjointness, and neutrality is automatic. The Gaussian field tails and a telescopic product estimate give

$$\left\| h_R - h_R^{(r)} \right\| \leq C \int_{d(0,S)>r/2} |\mathcal{V}_L(S)| \, \mathrm{d}S + C e^{-cr^2} \, \|\mathcal{V}_L\|_1 .$$

This proves (7). This sufficient condition does not cover the bare Coulomb potential $V(S) \propto |S|^{-1}$ in two spatial dimensions: its large-distance L^1 norm is not uniform in volume. An integrable potential with an algebraic tail $|V(S)| \sim |S|^{-\alpha}$, $\alpha > 2$, gives only a bound of order $r^{2-\alpha}$ by this construction. No extension to a fixed algebraic decay rate is claimed here.

Assumption 2.4 (Isolated low-energy band)**.** In the selected all-LLL number sector, list the eigenvalues of H_{LLL} as $E_1(L) \leq E_2(L) \leq \cdots$, counted with multiplicity. There are size-independent constants $d \in \mathbb{N}$, $w \geq 0$, and $\gamma > w$ such that

$$w_L := E_d(L) - E_1(L) \leq w, \qquad \gamma_L := E_{d+1}(L) - E_d(L) \geq \gamma. \tag{8}$$

Let P be the rank-d spectral projection onto these lowest d states. The positive gap above E_d ensures that this projection includes complete eigenspaces. If the all-LLL sector itself has dimension d, use $E_{d+1} = +\infty$.

We call Ran P the low-energy band or sector. In particular, d is not assumed to be the multiplicity of E_1. The condition $\gamma > w$ separates intraband energy differences from transitions across the gap. It suffices for the filter used in Appendix C. Assumption 1 of [2] uses the stronger condition $\gamma > 2w$ in our notation; we do not need that extra factor of two. If $w_L \to 0$ and the gap above the band stays positive, it holds for all sufficiently large sizes after choosing a fixed $w < \gamma$. Exponential decay of w_L is more than is needed. An exactly degenerate ground space is the special case $w_L = 0$.

2.4 An extensive penalty and its periodic dynamics

Boundedness of the density and the area $2\pi M$ give

$$\|H\| \le C_H M \tag{9}$$

with a size-independent C_H. Define

$$N_0 = \sum_\sigma \mathrm{d}\Gamma_\sigma(\Pi), \qquad N_{\mathrm{out}} = \sum_\sigma \mathrm{d}\Gamma_\sigma(1-\Pi), \qquad N = N_\uparrow + N_\downarrow = N_0 + N_{\mathrm{out}},$$

and introduce the auxiliary Hamiltonian

$$\mu_M = 2C_H M + \gamma + 1, \qquad H_{\mathrm{ext}} = H + \mu_M N_{\mathrm{out}}. \tag{10}$$

It is self-adjoint on the domain of N_{out}. The penalty vanishes on all-LLL states and is used only in the proof.

Lemma 2.5 (Spectral-band preservation)**.** *In either a fixed-spin or a fixed-total-number sector satisfying Assumption 2.4, H_{ext} has the same lowest d states, with the same energies, band projection P, and band width as H on the all-LLL sector. Its gap above this band is at least γ. No assumption on energies in smaller number sectors or on the density of empty orbitals is needed.*

Proof. The canonical extension of H commutes with both spin-resolved spectator numbers. Every block with at least one spectator particle has energy at least $-C_H M + \mu_M$, while the top all-LLL band energy obeys $E_d \le C_H M$. Hence the gap from E_d to any spectator block is at least

$$\mu_M - 2C_H M = \gamma + 1.$$

The no-spectator block is unchanged and its remaining spectrum lies at least γ above E_d. The same norm bound applies to all spin populations, proving both versions, including saturated sectors and the case that the entire all-LLL sector is the selected band. □

For neutral observables the dynamics of H_{ext} agrees with that of the bounded ambient operator

$$H_b = H - \mu_M N_0, \qquad H_{\mathrm{ext}} = H_b + \mu_M N. \tag{11}$$

On a fixed-total-number sector their spectra differ by the scalar $\mu_M N$, so their band projections, widths, and gaps agree. Furthermore,

$$[H, N_0] = 0, \qquad \mathrm{Spec}(N_0) \subset \{0, 1, \dots, 2M\}, \qquad \|H_b\| \le C_H M + 2\mu_M M = O(L^4).$$

Writing $\tau_t^G(A) = e^{\mathrm{i}tG} A e^{-\mathrm{i}tG}$, we have the exact factorization

$$\tau_t^{H_b} = \tau_t^H \circ \tau_{s_M(t)}^{N_0}, \qquad s_M(t) \equiv -\mu_M t \pmod{2\pi}, \qquad |s_M(t)| \le \pi. \tag{12}$$

The LLL kernel resolution gives

$$N_0 = \int_{\mathbb{T}_L} n_R \, \mathrm{d}R, \qquad n_R = \sum_\sigma a_\sigma^\dagger(e_R) a_\sigma(e_R), \qquad \|n_R\| \le C.$$

Restricting e_R to a ball produces self-adjoint neutral approximants to n_R with Gaussian errors. Thus the second factor in (12) has uniform locality estimates for its bounded time interval. Appendix B combines these estimates with those for H, rather than applying a generic Lieb–Robinson bound to the density of H_b, whose norm grows with M. The reduced time $s_M(t)$ may jump; periodicity makes the exact evolution continuous, and we never differentiate the reduced-time function.

Remark 2.6 (Why the physical charge is retained). On the LLL, the compression of Q_X is $\mathrm{d}\Gamma_\uparrow(\Pi 1_X \Pi)$. In general $(\Pi 1_X \Pi)^2 \neq \Pi 1_X \Pi$, so its spectrum need not be integral. We do not use this compressed charge in the flux exponentials. The physical charge Q_X can take a low-energy vector out of the LLL; the penalty makes these excursions compatible with a gap in the enlarged sector.

2.5 Verification from an isolated band at the original boundary condition

Proposition 2.7 (The analytical input)**.** *Let H satisfy Definition 2.3, and suppose its lowest d states in the selected all-LLL spin-number sector satisfy Assumption 2.4, with projection P. Use μ_M from (10) and H_b from (11). Then:*

1. *For every fixed $\eta > 0$ and bounded even ambient operator A supported in X, there are approximants $A_{t,\eta}$ supported in the physical $2\eta L$-neighborhood of X, with*

$$\sup_{|t|\le L^{1/3}} \left\| \tau_t^{H_b}(A) - A_{t,\eta} \right\| \le \|A\|\, O(L^{-\infty}), \qquad \|A_{t,\eta}\| = \|A\|\,.$$

   The constants are uniform in X, A, t. The approximation preserves adjoints and is neutral whenever A is neutral. In particular, bounded even observables separated by a distance proportional to L have commutator bounded by the product of their norms times $O(L^{-\infty})$ on this time interval.

2. *For $Q = Q_\Gamma$, the current is bounded on ambient Fock space and splits as*

$$\mathrm{i}[H_b, Q] = J_- + J_+.$$

   The currents have norm $O(ML_y) = O(L^3)$ and local approximants in boundary strips of radius r, with errors at most $C_k ML_y(1+r)^{-k}$ for every k.

3. *There are bounded, self-adjoint, neutral boundary dressings $K_\pm$, supported in strips of radius $L_x/64$, with polynomial ambient norms, such that*

$$[Q - K_- - K_+, P] \simeq 0.$$

4. *The low-energy-sector clustering estimate (4) holds for bounded neutral ambient operators supported in S_-, S_+.*

Proof. Appendices B and C prove these statements. For the first assertion, use Lemma B.2 with a smaller fixed cell buffer; comparison of physical and graph distances absorbs the bounded cell padding into $2\eta L$. Lemma 2.5 preserves the band and the lower gap bound $\Delta = \gamma > w$ in the enlarged sector. The densities, local approximants, currents, and dynamics preserve both spin numbers, so spectral filters of neutral operators connect only states in the selected sector. A grand-canonical gap is unnecessary. The inverse-energy filter handles transitions between P and its complement. The clustering filter vanishes on all intraband differences, whose magnitudes are at most w. Thus its projection identities remain exact when the band is split. Ambient norm bounds supply locality; the sector estimate $\|Q_X|_{\mathcal{H}_n}\| \le CL^2$ controls algebraic errors involving charges. The larger polynomial current norms do not change the $O(L^{-\infty})$ error class used in the index proof. □

For a unitary moving physical supports by a uniformly bounded distance, such as the magnetic translation below, conjugating these $K_\pm$ gives the $K_\pm^U$ required in Assumption 2.1. A unitary with rapidly decaying locality tails works as well if its neutral local approximants have uniform polynomial bounds.

3 Examples and the filling constraint

3.1 Magnetic translation

Let $\delta = L_x/M = 2\pi/L_y$, and define the one-particle magnetic translation

$$(t_x f)(x, y) = e^{\mathrm{i}\delta y} f(x - \delta, y).$$

It respects (1), commutes with Π, and moves spatial supports by δ. Write $\Gamma_{\mathrm{a}}(v) = \bigoplus_{k\geq 0} \wedge^k v$ for fermionic second quantization, distinct from the region Γ. Take $U = \Gamma_{\mathrm{a}}(t_x \otimes 1_{\mathbb{C}^2})^\dagger$, so that

$$U^\dagger Q_X U = Q_{X+\delta\hat{x}}. \tag{13}$$

Assume the interaction is invariant under this common translation of both spins. Then $[U, H_b] = [U, H_{\mathrm{ext}}] = [U, P] = 0$.

For $M > 2$, let

$$S^0_- = [0, \delta) \times S^1_y, \qquad S^0_+ = [L_x/2, L_x/2 + \delta) \times S^1_y.$$

Equation (13) gives exactly

$$U^\dagger Q U - Q = T_- + T_+, \qquad T_- = -Q_{S^0_-}, \quad T_+ = Q_{S^0_+}.$$

For large L these strips lie in $B_\pm$. Also $Q_- + T_-$ counts the region obtained by removing S^0_- from $\Gamma \cap S_-$, and $Q_+ + T_+$ counts the region obtained by adding S^0_+ to $\Gamma \cap S_+$. Thus (3) is exact. Their sector norms are at most $n_\uparrow$.

Theorem 3.1 (LLL filling constraint)**.** *Let the Hamiltonians, selected spin-number sectors, and low-energy band projections satisfy the hypotheses of Proposition 2.7, with fixed band rank d, and assume invariance under common magnetic translation. Then, for $\sigma = \uparrow, \downarrow$,*

$$\operatorname{dist}\left(\frac{dn_\sigma}{M}, \mathbb{Z}\right) = O(L^{-\infty}). \tag{14}$$

No spin-rotation symmetry or equality of the two populations is needed for this statement. For all sufficiently large tori these constraints imply the exact divisibility

$$M \mid dn_\uparrow, \qquad M \mid dn_\downarrow, \qquad \textit{equivalently} \qquad \frac{M}{\gcd(M, n_\uparrow, n_\downarrow)} \mid d. \tag{15}$$

If a spin filling n_σ/M converges, it equals its limit exactly for all sufficiently large sizes.

Proof. Proposition 2.7 and the exact transport calculation verify Assumption 2.1. The M translates of S^0_- partition the torus. Since $UPU^\dagger = P$, their charge traces are equal, and hence

$$M \operatorname{Tr}(P Q_{S^0_-}) = \operatorname{Tr}(P N_\uparrow) = dn_\uparrow.$$

Insert $\operatorname{Tr}(PT_-) = -dn_\uparrow/M$ into Theorem 2.2. Replacing the spin-up charge throughout by the spin-down charge proves the second assertion. For the final statement, a noninteger dn_σ/M is at distance at least $1/M$ from the integers. Since $M = O(L^2)$, this is incompatible with (14) for large L. The equivalent greatest-common-divisor formulation follows from Bézout's identity applied to $M, n_\uparrow, n_\downarrow$. Moreover, since $0 \leq n_\sigma \leq M$, the spin filling eventually belongs to the finite set $\{0, 1/d, \ldots, 1\}$. A convergent sequence in this set is eventually equal to its limit. □

3.2 Rational fillings and the parity distinction

Corollary 3.2 (Separate spin fillings)**.** *Under the hypotheses of Theorem 3.1, suppose*

$$\frac{n_\sigma}{M} \longrightarrow \nu_\sigma = \frac{a_\sigma}{b_\sigma}, \qquad \gcd(a_\sigma, b_\sigma) = 1, \quad b_\sigma > 0.$$

Then

$$\operatorname{lcm}(b_\uparrow, b_\downarrow) \mid d.$$

In particular, the conclusion holds when the two rational fillings are exactly fixed at every size.

Proof. The eventual equality $n_\sigma/M = \nu_\sigma$ and (15) give $d\nu_\sigma \in \mathbb{Z}$. Coprimality gives $b_\sigma \mid d$ for each spin, and hence the asserted least common multiple. The two constraints do not in general multiply: when the spin fillings coincide, they are the same arithmetic condition. □

Corollary 3.3 (Balanced rational filling)**.** *Under the hypotheses of Theorem 3.1, suppose*

$$n_\uparrow = n_\downarrow = N/2, \qquad \frac{N}{M} \longrightarrow \nu = \frac{p}{q}, \qquad 0 < \nu \le 2, \quad p, q \in \mathbb{N}, \quad \gcd(p, q) = 1.$$

Then the low-energy band rank in this balanced sector obeys

$$\boxed{b_\nu \mid d, \qquad b_\nu := \frac{2q}{\gcd(p, 2q)} = \frac{2q}{\gcd(p, 2)} = \begin{cases} 2q, & p \text{ odd}, \\ q, & p \text{ even}. \end{cases}} \tag{16}$$

Proof. The spin filling is $\nu_\sigma = p/(2q)$. Applying Corollary 3.2, or taking the limit directly in (14), gives

$$\frac{dp}{2q} \in \mathbb{Z}. \tag{17}$$

The denominator of $p/(2q)$ in lowest terms is $2q/\gcd(p, 2q)$. Since p and q are coprime, $\gcd(p, 2q) = \gcd(p, 2)$, proving the result. □

This corollary applies to the bounded, rapidly local LLL interactions of Definition 2.3, including the endpoint $\nu = 2$, with an isolated low-energy band satisfying Assumption 2.4. The lowest energy is arbitrary and splitting inside the band is allowed. The extensive auxiliary penalty preserves the band without a particle-removal estimate or a positive density of empty orbitals.

The parity distinction can equivalently be organized by the denominator of the total filling:

q	p (coprime to q)	divisor forced on d
even	necessarily odd	$2q$
odd	odd	$2q$
odd	even	q

Thus even q always gives a multiple of $2q$. For odd q, the parity of p decides whether the factor of two survives. Section 3.3 compares examples with the exact translation degeneracy. These are necessary divisors; the theorem does not assert the existence of a gapped state attaining any of these dimensions.

There is no second factor of two from applying the theorem to spin down: at balance it reproduces (17). Using total charge gives $dp/q \in \mathbb{Z}$, which is already implied by that equation. More generally, integer combinations of the two spin charges give integer combinations of the same two transport indices and do not strengthen this arithmetic.

Remark 3.4 (Admissible finite tori). If $N/M = p/q$ is imposed exactly and the populations are balanced, the allowed sizes are

$$M = b_\nu k, \qquad n_\uparrow = n_\downarrow = \frac{p}{\gcd(p,2)}k, \qquad k \in \mathbb{N}.$$

Indeed, integrality of $n_\sigma = Mp/(2q)$ is equivalent to $b_\nu \mid M$. In particular, when p is odd one must take an even multiple of q flux quanta to have even N. This finite-size restriction does not supply another factor in the low-energy band rank. The limiting-filling formulation does not assume exact rational filling at each finite size. Under the band hypotheses, Theorem 3.1 shows that convergence of a spin filling forces eventual equality to its limit.

3.3 Comparison with exact translation degeneracy

For this comparison, impose symmetry also under the common y-translation. The additional one-particle unitary is

$$(t_y f)(x,y) = f(x, y-\delta_y), \qquad \delta_y = L_y/M.$$

It respects (1) because $L_x\delta_y = 2\pi$, and it commutes with the covariant derivatives and Π. Direct substitution gives

$$t_x t_y = e^{\mathrm{i}\delta\delta_y} t_y t_x = e^{2\pi\mathrm{i}/M} t_y t_x.$$

Set $\mathcal{T}_\alpha = \Gamma_{\mathrm{a}}(t_\alpha \otimes 1_{\mathbb{C}^2})$, for $\alpha = x, y$. Thus $\mathcal{T}_x = U^\dagger$ in the convention used above. On a fixed-total-number sector, and also on any fixed-spin-number sector with total number N, their algebra is

$$\mathcal{T}_x \mathcal{T}_y = e^{2\pi\mathrm{i}N/M} \mathcal{T}_y \mathcal{T}_x. \tag{18}$$

Proposition 3.5 (Exact center-of-mass degeneracy)**.** *If an LLL Hamiltonian commutes with both $\mathcal{T}_x$ and $\mathcal{T}_y$, the dimension d_E of each energy eigenspace in either of these number sectors satisfies*

$$q_L := \frac{M}{\gcd(M,N)} \mid d_E.$$

This is a finite-size symmetry statement, requiring neither a gap nor locality nor positivity of the interaction.

Proof. The eigenspace is finite-dimensional and invariant under the two translations. Take the determinant of (18) restricted to it. Cancelling the determinants of the unitaries gives $e^{2\pi\mathrm{i}Nd_E/M} = 1$, or $M \mid Nd_E$. This is equivalent to the stated divisibility. □

This is the center-of-mass part of Haldane's magnetic-translation analysis [4]. An isolated band contains complete eigenspaces, so its rank also obeys $q_L \mid d$. At exact total filling $N/M = p/q$ in lowest terms, $q_L = q$. The main index theorem requires only the common x-translation; when both translations are symmetries, its improvement can be quantified. For an isolated low-energy band as in Assumption 2.4 in a fixed spin-number sector, (15) gives, for all sufficiently large sizes,

$$b_L := \frac{M}{\gcd(M, n_\uparrow, n_\downarrow)} \mid d, \qquad \frac{b_L}{q_L} = \frac{\gcd(M,N)}{\gcd(M, n_\uparrow, n_\downarrow)}.$$

The quotient is an integer because a common divisor of $M, n_\uparrow, n_\downarrow$ also divides $N = n_\uparrow + n_\downarrow$. At balanced exact filling, $b_L = b_\nu$, so the gain is $2/\gcd(p,2)$: a factor of two for odd p, and no gain in this divisibility constraint for even p. For a fully polarized sector, say $n_\downarrow = 0$, one has $b_L = q_L$ as well.

The proof of Proposition 3.5 is included here so that this comparison does not require a result from the cited paper. The two conclusions concern different spectral objects: the translation algebra constrains each exact energy eigenspace, whereas the index constrains the whole isolated band. For example, at $\nu = 1/2$, a rank-four low-energy band split into two translation doublets is compatible with both conclusions.

3.4 The full low-energy band with spin-exchange symmetry

Specifying spin-1/2 particles alone does not imply balanced populations. One may impose the balanced sector as above. Alternatively, a spin symmetry gives precisely the trace identity needed for the full low-energy band, without requiring every vector in it to be a singlet.

Corollary 3.6 (Spin-resolved band)**.** *Let H_{LLL} satisfy Definition 2.3 and be invariant under common magnetic translation. In the full all-LLL sector of total number N, suppose its lowest d states satisfy Assumption 2.4, with projection P. No spin-exchange symmetry is required. Since P commutes with $N_\uparrow$, write*

$$P = \sum_{n_\uparrow + n_\downarrow = N} P_{\boldsymbol{n}}, \qquad P_{\boldsymbol{n}} = P\,\mathbf{1}_{\{n_\uparrow\}}(N_\uparrow), \qquad d_{\boldsymbol{n}} = \operatorname{rank} P_{\boldsymbol{n}}.$$

Then, for all sufficiently large tori, every sector with $d_{\boldsymbol{n}} > 0$ satisfies

$$M \mid d_{\boldsymbol{n}} n_\uparrow, \qquad M \mid d_{\boldsymbol{n}} n_\downarrow, \qquad \textit{equivalently} \qquad \frac{M}{\gcd(M, n_\uparrow, n_\downarrow)} \mid d_{\boldsymbol{n}}. \tag{19}$$

In particular, $d_{\boldsymbol{n}} = 0$ whenever $0 < |n_\uparrow - n_\downarrow| < M/d$.

Proof. Let $E_1 \le E_2 \le \cdots$ be the energies of the fixed-N all-LLL sector, with band constants w and $\gamma > w$.

The density in Definition 2.3 is neutral in each spin, so H_{LLL} is reduced by the fixed-spin parts of the all-LLL fixed-N sector. Hence P commutes with $N_\uparrow$, and $P_{\boldsymbol{n}}$ is the spectral projection of H_{LLL} on the all-LLL part of $\mathcal{H}_{\boldsymbol{n}}$ for the interval $[E_1, E_d]$. All other eigenvalues in that part are at least E_{d+1}.

Fix a sector with $d_{\boldsymbol{n}} > 0$. Its lowest energy $E_0(\boldsymbol{n})$ lies in $[E_1, E_d]$, so $P_{\boldsymbol{n}}$ projects onto its lowest $d_{\boldsymbol{n}}$ states. Their width is at most $E_d - E_1 \le w$, and the next eigenvalue in the sector is at least $E_{d+1} \ge E_d + \gamma$. Thus Assumption 2.4 holds in this sector, with rank $d_{\boldsymbol{n}}$ and the same w and γ. Lemma 2.5 preserves this band using the same μ_M in every spin sector: its norm comparison is uniform in the populations, including saturated ones.

Suppose the first assertion fails for infinitely many sizes. For each such size choose a sector $\boldsymbol{n}(L)$ with $d_{\boldsymbol{n}(L)} > 0$ that violates (19). Since $1 \le d_{\boldsymbol{n}(L)} \le d$, a further subsequence has constant rank d'. Along it, the previous paragraph verifies the hypotheses of Proposition 2.7 for the sectors $\boldsymbol{n}(L)$ and projections $P_{\boldsymbol{n}(L)}$; invariance under common magnetic translation is assumed. Theorem 3.1, applied along this subsequence, gives $M \mid d' n_\uparrow(L)$ and $M \mid d' n_\downarrow(L)$ for all large sizes in it, a contradiction. The greatest-common-divisor form is the equivalence in (15).

Finally, $\gcd(M, n_\uparrow, n_\downarrow)$ divides $n_\uparrow - n_\downarrow$. If $d_{\boldsymbol{n}} > 0$ and $n_\uparrow \ne n_\downarrow$, then (19) gives $M/|n_\uparrow - n_\downarrow| \le M/\gcd(M, n_\uparrow, n_\downarrow) \le d_{\boldsymbol{n}} \le d$. □

Theorem 3.7 (Spin-symmetric fixed-total-number version)**.** *Let the Hamiltonians of Definition 2.3 be invariant under common magnetic translation. In the full all-LLL sector of fixed total particle number $N = N_\uparrow + N_\downarrow$, suppose the lowest d states satisfy Assumption 2.4: d is fixed, their width is at most w, and the gap above them is at least $\gamma > w$. Let P be the spectral projection onto this*

band, including all its spin sectors. Assume an ambient unitary $\mathcal{R}$ preserves the all-LLL subspace, commutes with the interaction, and exchanges the two spin numbers:

$$[\mathcal{R}, H_{\mathrm{LLL}}] = 0,$$

$$\mathcal{R}^\dagger N_\uparrow \mathcal{R} = N_\downarrow, \qquad \mathcal{R}^\dagger N_\downarrow \mathcal{R} = N_\uparrow.$$

Then

$$\operatorname{dist}\left(\frac{dN}{2M}, \mathbb{Z}\right) = O(L^{-\infty}). \tag{20}$$

In particular, $2M \mid dN$ for all sufficiently large tori. If $N/M \to p/q$ in lowest terms, the same divisor $b_\nu = 2q/\gcd(p,2)$ in (16) divides this full low-energy band rank. Charge conservation and SU(2) *spin symmetry imply the spin conservation and exchange assumptions used here.*

Proof. Corollary 3.6 gives, for all sufficiently large sizes,

$$\frac{1}{M}\operatorname{Tr}(PN_\uparrow) = \sum_{\boldsymbol{n}} \frac{d_{\boldsymbol{n}} n_\uparrow}{M} \in \mathbb{Z}.$$

Spin exchange preserves total number. Thus $\mathcal{R}$ acts unitarily on the all-LLL fixed-N sector and commutes there with H_{LLL}, hence with its spectral projection P. Consequently $\mathcal{R}^\dagger P \mathcal{R} = P$, and cyclicity gives

$$\operatorname{Tr}(PN_\uparrow) = \operatorname{Tr}(PN_\downarrow), \qquad \operatorname{Tr}(P(N_\uparrow + N_\downarrow)) = dN.$$

Each spin trace is therefore $dN/2$, so $dN/(2M)$ is an integer for all sufficiently large sizes. This proves (20) and $2M \mid dN$. If $N/M \to p/q$, taking the limit gives $dp/(2q) \in \mathbb{Z}$, and the arithmetic of Corollary 3.3 gives $b_\nu \mid d$. For an SU(2)-invariant Hamiltonian, the second quantization of a rotation by π about the spin x-axis supplies $\mathcal{R}$. It acts only on spin, so it preserves the all-LLL subspace. Conservation of N and S^z gives conservation of both spin numbers. □

For this theorem the statement $\nu_\uparrow = \nu_\downarrow = \nu/2$ means the filling in the normalized band trace:

$$\frac{\operatorname{Tr}(PN_\sigma)}{dM} = \frac{N}{2M}.$$

It need not hold as a sharp particle number in each vector in the band, and no evenness assumption on N is made; if exact filling is imposed, $M = qk$, $N = pk$ suffices. By contrast, Corollary 3.3 concerns the balanced sector alone and requires a gap only within that sector. The two ranks agree when the full fixed-N band lies in the balanced sector. Corollary 3.6 shows that a band of bounded rank can leave the balanced sector only through sectors with $|n_\uparrow - n_\downarrow| \geq M/d$. Corollary 3.8 shows that with SU(2) symmetry it does not leave it at all on large tori; in particular N is then even.

Corollary 3.8 (Spin-singlet bands)**.** *In Theorem 3.7, suppose H_{LLL} is* SU(2)*-invariant. Then, for all sufficiently large tori, every vector in* $\operatorname{Ran} P$ *has total spin zero. In particular N is even,* $\operatorname{Ran} P$ *lies in the balanced sector $n_\uparrow = n_\downarrow = N/2$, and P is the low-energy band projection of that sector for the same energy window, with the same rank, width bound, and gap.*

Proof. The total spin operators $\boldsymbol{S} = \mathrm{d}\Gamma(1_{\mathfrak{h}} \otimes \boldsymbol{\sigma}/2)$ commute with H_{LLL} and preserve the all-LLL fixed-N sector, so they commute with its spectral projection P. Thus $\operatorname{Ran} P$ is a finite-dimensional SU(2)-representation and splits into irreducible multiplets. Suppose one of them has spin $S \geq 1/2$. It contains a vector with $S^z = 1$ if S is an integer, and with $S^z = 1/2$ otherwise. Since $2S^z = N_\uparrow - N_\downarrow$, some sector with $n_\uparrow - n_\downarrow \in \{1, 2\}$ has $d_{\boldsymbol{n}} > 0$. Once $M > 2d$, this contradicts the last assertion of

Corollary 3.6. Hence every multiplet in $\operatorname{Ran} P$ is a singlet. A singlet has $S^z = 0$, so it lies in the sector $n_\uparrow = n_\downarrow = N/2$.

In that sector, $\operatorname{Ran} P$ contains every state of energy at most E_d, and the remaining states have energy at least E_{d+1}. So P is the band projection of the balanced sector, with the same constants. □

Under the full-sector band hypotheses and SU(2) symmetry, therefore, the fixed-total-number band and the balanced-sector band of Corollary 3.3 coincide on large tori. The distinction between the two formulations matters only for spin-exchange symmetry without full SU(2), where Corollary 3.6 still allows extensively polarized sectors with $|n_\uparrow - n_\downarrow| \geq M/d$.

Corollary 3.9 (Non-singlet low-energy states)**.** *Let H_{LLL} satisfy Definition 2.3, and let it be* SU(2)*-invariant and invariant under common magnetic translation. In the all-LLL sectors of total number N, fix d, and suppose $w_L = E_d(L) - E_1(L) \to 0$. Suppose also that, for all sufficiently large tori, $\boldsymbol{S}^2$ does not vanish on the spectral subspace of eigenvalues at most $E_d(L)$. Then*

$$E_{d+1}(L) - E_d(L) \longrightarrow 0.$$

Proof. Otherwise some subsequence has $E_{d+1} - E_d \geq \gamma_0 > 0$. Choose a fixed $w < \gamma_0$ bounding w_L eventually along it. Assumption 2.4 then holds there, and P is the spectral projection for eigenvalues at most E_d. The SU(2) symmetry supplies the exchange unitary of Theorem 3.7. Corollary 3.8, applied along the subsequence, gives $\boldsymbol{S}^2 P = 0$ for large sizes, a contradiction. □

For example, an exactly degenerate lowest multiplet of spin $S \geq 1/2$ has $d = 2S + 1$ and $w_L = 0$; if S stays bounded, the gap above it closes. For odd N every state has half-integer spin, so an SU(2)-invariant Hamiltonian in this class has no uniformly gapped low-energy band of bounded rank at odd particle number. This is analogous to a Lieb–Schultz–Mattis statement. It does not apply to a fully polarized lowest multiplet with $S = N/2$, whose dimension $N + 1$ is unbounded.

3.5 Special cases and the gap obstruction

Corollary 3.10 (Balanced half filling)**.** *At $\nu = 1/2$, either the balanced-sector hypotheses of Corollary 3.3 or the full-sector hypotheses of Theorem 3.7 imply $4 \mid d$.*

Proof. Here $p = 1$, $q = 2$, and hence $b_\nu = 4$. □

Corollary 3.11 (Band-gap obstruction)**.** *Fix a rational limiting filling $\nu = p/q$ and either the balanced-sector or the full spin-exchange-symmetric setting. Retain the boundedness, rapid physical locality, and symmetry hypotheses, but do not assume a uniform spectral gap. For a fixed d with $b_\nu \nmid d$, suppose the widths $w_L = E_d(L) - E_1(L)$ are uniformly bounded, and write $\gamma_L = E_{d+1}(L) - E_d(L)$. Then*

$$\limsup_{L \to \infty} (\gamma_L - w_L) \leq 0. \tag{21}$$

In particular, if $w_L \to 0$, then

$$E_{d+1}(L) - E_d(L) \longrightarrow 0, \qquad E_{d+1}(L) - E_1(L) \longrightarrow 0. \tag{22}$$

Proof. If (21) failed, some subsequence would satisfy $\gamma_L - w_L \geq \delta > 0$. By boundedness, pass to a further subsequence with $w_L \to w_* \geq 0$. Eventually on it,

$$w_L \leq w_* + \delta/4 =: w, \qquad \gamma_L \geq w_* + 3\delta/4 \geq \gamma := w_* + \delta/2 > w.$$

Assumption 2.4 then holds with fixed constants, and the corresponding divisibility theorem gives $b_\nu \mid d$, a contradiction. If $w_L \to 0$, nonnegativity of γ_L gives $\gamma_L \to 0$. Adding w_L proves the second limit in (22). □

At balanced half filling, a Hamiltonian in the stated class with exactly ten ground states at any common energy $E_1(L)$ has $d = 10$, $w_L = 0$, and hence $E_{11}(L) - E_1(L) \to 0$. For zero-energy ground states this reads $E_{11}(L) \to 0$. No sign assumption or energy comparison with other particle-number sectors is required. The same obstruction applies to a proposed rank-ten band with exponentially small, or merely vanishing, splitting. Spectral counting is a separate input; no assertion about that count is used or proved in this note.

It is essential to distinguish these two uses of d. Ten exact zero modes alone do not exclude a larger isolated low-energy band: additional positive-energy states could tend to zero and produce a band of rank twelve, sixteen, or another multiple of four. The conclusion from the exact zero-mode count alone is $E_{11} \to 0$. To exclude a uniformly gapped phase with a finite quasi-degenerate low-energy sector by this argument, one must identify the rank of that entire sector and show that it violates the divisor, or otherwise exclude all such larger bands.

When both common magnetic translations are symmetries, the exact algebra at $\nu = 1/2$ requires only $2 \mid d$, which is compatible with ten zero modes. The additional band constraint $4 \mid d$ contradicts a uniform gap above a rank-ten band whose width vanishes. This identifies both the gap assumption used for contradiction and the spectral sector to which it applies.

The distinction for even p is necessary even for a spin-singlet ground state. For example, at $\nu = 2$ the two spin components of the LLL are completely filled. The all-LLL sector with $N = 2M$ is one-dimensional and its filled state is a spin singlet. Take $H_{\mathrm{LLL}} = 0$, which belongs to Definition 2.3. After the enlargement its Hamiltonian is $\mu_M \sum_\sigma \mathrm{d}\Gamma_\sigma(1 - \Pi)$, with this unique ground state and gap μ_M at fixed N. The all-LLL sector itself has no orthogonal states. Thus a universal claim $2q \mid d$ would already fail at $p = 2, q = 1, d = 1$, whereas (16) gives the correct divisor one.

4 Proof of the index theorem

We work on $\mathcal{H}_{\boldsymbol{n}}$ when manipulating charges, and continue to use ambient supports for local unitaries. On this sector all charges and their bounded perturbations have polynomial norms. The proof uses the rank and approximate invariance of P, the dressing identity, and clustering. It never requires the Hamiltonian to be a scalar on $\operatorname{Ran} P$. Every error below is uniform for $0 \le \phi \le 2\pi$.

4.1 Approximate invariant subspaces

Lemma 4.1 (Separated factors)**.** *Let $V_\pm$ be neutral ambient unitaries supported in $S_\pm$. If $[V_- V_+, P] \simeq 0$ and (4) holds, then $[V_\pm, P] \simeq 0$.*

Proof. Write $V = V_+ V_-$; the factors commute. For every unit vector $\psi \in \operatorname{Ran} P$, approximate invariance gives $\|PV\psi\| \ge 1 - O(L^{-\infty})$. Clustering gives

$$PV_+V_-P \simeq PV_+PV_-P.$$

Here the order of supports is reversed relative to (4); the required estimate follows from

$$\|PV_+(1-P)V_-P\| = \left\|PV_-^\dagger(1-P)V_+^\dagger P\right\| = O(L^{-\infty}).$$

Since $\|PV_+P\| \le 1$, it follows that $\|PV_-\psi\| \ge 1 - O(L^{-\infty})$. Consequently $\|(1-P)V_-P\| = O(L^{-\infty})$. Apply the same argument to $V^\dagger$, reordering its separated factors, to get $\left\|(1-P)V_-^\dagger P\right\| = O(L^{-\infty})$.

These are the two off-diagonal blocks of $[V_-, P]$. Interchanging the two signs proves the other assertion. □

Lemma 4.2 (Compression of an almost invariant unitary)**.** *If V is unitary and $[V, P] \simeq 0$, then $PVP|_{\operatorname{Ran} P}$ differs by $O(L^{-\infty})$ from a unitary on* $\operatorname{Ran} P$*. If $A = A^\dagger$ is sector-bounded and $[A, P] \simeq 0$, then $[e^{\mathrm{i}\phi A}, P] \simeq 0$.*

Proof. For $v = PVP|_{\operatorname{Ran} P}$,

$$v^\dagger v = P - PV^\dagger(1-P)VP \simeq P.$$

Its polar factor $v(v^\dagger v)^{-1/2}$ gives the first claim. For the second use

$$[e^{\mathrm{i}\phi A}, P] = \mathrm{i}\int_0^\phi e^{\mathrm{i}sA}[A, P]e^{\mathrm{i}(\phi-s)A}\,\mathrm{d}s.$$

The unitaries have norm one, so the estimate does not grow exponentially with $\|A\|$. □

4.2 Local charge fluctuations

Define

$$\bar{Q}_\pm = Q_\pm - K_\pm, \quad \bar{Q} = Q - K_- - K_+ = \bar{Q}_- + Q_m + \bar{Q}_+,$$

and

$$\bar{Q}^U_\pm = Q_\pm + T_\pm - K^U_\pm, \qquad \bar{Q}^U = \bar{Q}^U_- + Q_m + \bar{Q}^U_+.$$

The support choices ensure that objects with opposite boundary signs commute and that Q_m commutes with all boundary objects. In particular,

$$\bar{Q}^U \simeq U^\dagger \bar{Q} U, \qquad [\bar{Q}, P] \simeq 0, \qquad [\bar{Q}^U, P] \simeq 0. \tag{23}$$

All of these statements are sector norm statements. Exponentials of the local operators also have ambient local realizations: their unbounded parts are the local charges in (3), and their perturbations are bounded.

Set

$$Z(\phi) = e^{\mathrm{i}\phi\bar{Q}^U}e^{-\mathrm{i}\phi\bar{Q}}, \qquad Z_\pm(\phi) = e^{\mathrm{i}\phi\bar{Q}^U_\pm}e^{-\mathrm{i}\phi\bar{Q}_\pm}. \tag{24}$$

The commuting spatial decomposition gives $Z = Z_-Z_+$. Equation (23) gives $[Z, P] \simeq 0$. Lemma 4.1 therefore implies

$$[Z_\pm(\phi), P] \simeq 0. \tag{25}$$

Similarly, since Q_m is integer-valued,

$$e^{2\pi\mathrm{i}\bar{Q}} = e^{2\pi\mathrm{i}\bar{Q}_-}e^{2\pi\mathrm{i}\bar{Q}_+}.$$

Another application of Lemma 4.1 yields

$$[e^{2\pi\mathrm{i}\bar{Q}_\pm}, P] \simeq 0. \tag{26}$$

4.3 A formula for the determinant

On the d-dimensional space $\operatorname{Ran} P$, set

$$\mathcal{Z}_-(\phi) = PZ_-(\phi)P, \qquad D_- = \bar{Q}_-^U - \bar{Q}_- = T_- - K_-^U + K_-.$$

By (25), $\mathcal{Z}_-$ is uniformly close to a unitary and hence invertible, with uniformly bounded inverse. Differentiating the bounded sector operators gives

$$\begin{aligned} -\mathrm{i}\mathcal{Z}_-'(\phi) &= PZ_-(\phi)e^{\mathrm{i}\phi\bar{Q}_-}D_-e^{-\mathrm{i}\phi\bar{Q}_-}P \\ &\simeq \mathcal{Z}_-(\phi)Pe^{\mathrm{i}\phi\bar{Q}}D_-e^{-\mathrm{i}\phi\bar{Q}}P. \end{aligned} \tag{27}$$

In the second line one first inserts P after Z_-, using (25); the remaining parts of $\bar{Q}$ commute with D_-. The polynomial norm of D_- does not change the error class. Since $[\bar{Q}, P] \simeq 0$, the trace of the last factor is $\operatorname{Tr}(PD_-) + O(L^{-\infty})$.

Jacobi's formula for the determinant now gives

$$\frac{\mathrm{d}}{\mathrm{d}\phi} \det \mathcal{Z}_-(\phi) = \left(\mathrm{i}\operatorname{Tr}(PD_-) + O(L^{-\infty})\right) \det \mathcal{Z}_-(\phi).$$

Because D_- is self-adjoint and $\mathcal{Z}_-(0) = 1_{\operatorname{Ran} P}$, integration over a bounded interval yields

$$\det \mathcal{Z}_-(\phi) = e^{\mathrm{i}\phi \operatorname{Tr}(PD_-)} + O(L^{-\infty}) = e^{\mathrm{i}\phi \operatorname{Tr}(PT_-)} + O(L^{-\infty}). \tag{28}$$

For the last equality use $K_-^U \simeq U^\dagger K_- U$ and $[U, P] \simeq 0$. Cyclicity of the finite-rank trace gives $\operatorname{Tr}(PK_-^U) - \operatorname{Tr}(PK_-) = O(L^{-\infty})$. This is the finite-rank counterpart of computing the phase acquired by a single invariant state.

4.4 The endpoint and integrality

Let $V = e^{2\pi\mathrm{i}\bar{Q}_-}$. The integer spectra of Q_m, Q_+ give

$$V = e^{2\pi\mathrm{i}(Q - K_-)}.$$

Conjugate by U, use the transport identity and $K_-^U \simeq U^\dagger K_- U$, and split the commuting boundary terms. The normalization $e^{2\pi\mathrm{i}(Q_+ + T_+)} = 1$ gives

$$U^\dagger V U \simeq e^{2\pi\mathrm{i}(Q + T_- + T_+ - K_-^U)} = e^{2\pi\mathrm{i}\bar{Q}_-^U}.$$

It follows that

$$Z_-(2\pi) \simeq U^\dagger V U V^\dagger.$$

Both U and V preserve P up to $O(L^{-\infty})$, by Assumption 2.1 and (26). After compression and replacement by their polar unitaries (Lemma 4.2), the right-hand side is a commutator of two d-dimensional unitaries, up to $O(L^{-\infty})$. Its determinant is therefore $1 + O(L^{-\infty})$. This is the same finite-dimensional determinant identity used in Proposition 3.5; here the second unitary is the dressed spin-charge rotation V, rather than the common translation $\mathcal{T}_y$. Equation (28) implies

$$e^{2\pi\mathrm{i}\operatorname{Tr}(PT_-)} = 1 + O(L^{-\infty}).$$

For real x, distance from x to $\mathbb{Z}$ is bounded by a constant times $\left|e^{2\pi\mathrm{i}x} - 1\right|$. This proves Theorem 2.2.

A Gaussian LLL kernels on the torus

Let $\kappa = 2\pi/L_y$. An orthonormal basis of the LLL is

$$\varphi_n(x,y) = (L_y\sqrt{\pi})^{-1/2} e^{\mathrm{i}\kappa n y} e^{-(x-\kappa n)^2/2}, \qquad \psi_j = \sum_{m\in\mathbb{Z}} \varphi_{j+mM}, \quad 0 \le j < M.$$

The sums and their derivatives converge locally uniformly. They satisfy (1) and the LLL equation. Integration in y kills unequal Fourier modes, and the remaining intervals in x tile the real line; hence the ψ_j are orthonormal. Conversely, Fourier expansion in y and the LLL equation force a Gaussian in x for each Fourier coefficient, while the x-boundary condition makes the coefficients periodic modulo M. This proves completeness.

The projector kernel is $\Pi(r,r') = \sum_j \psi_j(r)\overline{\psi_j(r')}$. Expanding the two sums and applying Poisson summation to the remaining Fourier label expresses it as a sum indexed by $(\ell_1,\ell_2) \in \mathbb{Z}^2$, whose summands have modulus

$$\frac{1}{2\pi} \exp\left[-\frac{(x-x'+\ell_1 L_x)^2 + (y-y'+\ell_2 L_y)^2}{4}\right].$$

For example, before Poisson summation the expression is

$$\frac{1}{L_y\sqrt{\pi}} \sum_{\ell_1\in\mathbb{Z}} e^{-\mathrm{i}\ell_1 L_x y'} e^{-(x-x'+\ell_1 L_x)^2/4} \sum_{n\in\mathbb{Z}} e^{\mathrm{i}\kappa n(y-y')} e^{-(\kappa n - (x+x'-\ell_1 L_x)/2)^2}.$$

The Poisson formula

$$\sum_n e^{-(\kappa n - a)^2} e^{\mathrm{i}\kappa n b} = \frac{\sqrt{\pi}}{\kappa} \sum_{\ell_2} e^{-(b+\ell_2 L_y)^2/4} e^{\mathrm{i}a(b+\ell_2 L_y)}$$

gives the claimed modulus term by term. Covariant derivatives of the kernel introduce polynomials in the image displacement. The connection terms cancel the coordinate-dependent part of the magnetic phase. In particular, for a fixed number s of covariant derivatives,

$$|\nabla^s_{\mathrm{cov}} \Pi(r,r')| \le C_s \sum_{\lambda \in L_x\mathbb{Z}\oplus L_y\mathbb{Z}} (1+|r-r'+\lambda|)^s e^{-|r-r'+\lambda|^2/4}. \tag{29}$$

The derivatives here have the appropriate dual connection in the conjugate variable.

For $L \ge 4$, the image lattice has uniform separation. If $a_\lambda = |r - R + \lambda|$, then $\sum_\lambda e^{-a_\lambda^2/16} \le C$. Cauchy–Schwarz, with part of the Gaussian as a weight, therefore bounds the square of the right side of (29) by $C_s \sum_\lambda e^{-a_\lambda^2/8}$, after absorbing polynomials into the Gaussian. On the set $d(r,R) > r_0$ all $a_\lambda > r_0$. Integrating over a fundamental domain and summing over images is bounded by the plane integral

$$C_s \int_{|z|>r_0} e^{-|z|^2/8}\, \mathrm{d}z \le C_s' e^{-cr_0^2}.$$

This proves the L^2 tail bounds for $e_R = \Pi(\cdot, R)$ and for g_R, which is obtained by differentiating the evaluation functional. Their uniform norm bounds follow by setting $r_0 = 0$. Finally, the orthonormal expansion gives

$$\int |e_R\rangle\langle e_R|\, \mathrm{d}R = \sum_{j,k} |\psi_j\rangle\langle\psi_k| \int \overline{\psi_j(R)}\psi_k(R)\, \mathrm{d}R = \Pi.$$

B Locality without a localized LLL basis

B.1 Spatial cells and a bounded interaction

Partition the physical torus into rectangles of side lengths between 1 and 2, and denote their adjacency graph by Λ_L. Its degree and quadratic volume-growth constants are uniform, and $V_L := |\Lambda_L| = O(L^2)$. Graph distance and physical distance are comparable up to uniform multiplicative and additive constants. We use graph distance in this appendix. Replacing a physical support by the cells it meets adds only a bounded spatial padding, which is absorbed in the macroscopic buffers below. A cell carries the infinite-dimensional Fock space of its L^2 sections; no basis in that space is chosen.

Write $H = H_{\mathrm{LLL}}$. Definition 2.3 supplies self-adjoint neutral approximants with

$$\|h_R\| \le C, \qquad \left\|h_R - h_R^{(r)}\right\| \le C_k(1+r)^{-k} \quad \text{for every } k. \tag{30}$$

For each cell z, put $A_z = \int_z h_R \,\mathrm{d}R$ and $A_z^{(r)} = \int_z h_R^{(r)} \,\mathrm{d}R$. Weak operator integrals suffice for the approximants: the norm estimates pass through the integrals, and local von Neumann algebras are weakly closed. Set

$$\Phi_{z,0} = A_z^{(1)}, \qquad \Phi_{z,n} = A_z^{(n+1)} - A_z^{(n)} \quad (n \ge 1).$$

Each term is self-adjoint and neutral, is supported on a cell set $Z_{z,n} \subset B_{C(n+1)}(z)$, and satisfies

$$\|\Phi_{z,n}\| \le C_k(1+n)^{-k} \quad \text{for every } k.$$

The approximants equal A_z once the radius covers the torus, so $H = \sum_{z,n} \Phi_{z,n}$. The sum is norm convergent. This is a decomposition by physical regions, not by LLL orbitals.

Collecting terms with the same support gives an interaction $\Phi(Z)$. For every fixed $p \ge 0$,

$$\sup_x \sum_{Z \ni x} |Z|(1+\operatorname{diam} Z)^p \,\|\Phi(Z)\| \le C_p. \tag{31}$$

Indeed, a shell of index n containing x has at most $C(1+n)^2$ possible centers, at most $C(1+n)^2$ cells, and diameter at most $C(1+n)$. Its contribution is bounded by $C_k(1+n)^{p+4-k}$, summable when $k > p+5$. In particular,

$$\sum_Z |Z| \,\|\Phi(Z)\| \le CV_L, \qquad \sup_{x,y}(1+d(x,y))^p \sum_{Z \ni x,y} \|\Phi(Z)\| \le C_p.$$

For $\ell \ge 1$, retain only shells with $n \le \lfloor \ell \rfloor$, and call the resulting Hamiltonian $H^{(\ell)}$ and interaction $\Phi^{(\ell)}$. Its support diameters are at most $C_{\mathrm{diam}}\ell$, with a uniform C_{diam}, and

$$\left\|H - H^{(\ell)}\right\| \le C_k V_L (1+\ell)^{-k} \quad \text{for every } k.$$

All preceding summability bounds also hold for the retained shells, whether or not collecting them causes cancellations. With

$$F_\ell(r) = \frac{e^{-r/\ell}}{(1+r)^4},$$

we consequently have

$$\mathcal{I}_\ell := \sup_{x,y} \frac{\sum_{Z \ni x,y} \left\|\Phi^{(\ell)}(Z)\right\|}{F_\ell(d(x,y))} \le C, \qquad \sum_z F_\ell(d(x,z))F_\ell(d(z,y)) \le C_F F_\ell(d(x,y)). \tag{32}$$

Both constants are independent of L and ℓ. For the first inequality, a nonzero summand has $d(x,y) \le C_{\mathrm{diam}}\ell$, so the exponential weight costs at most $e^{C_{\mathrm{diam}}}$, and the polynomial estimate with $p = 4$ applies. For the second, the triangle inequality bounds the quotient of exponential weights by one. Split the sum according to which of $d(x,z), d(z,y)$ is at least $d(x,y)/2$; the corresponding polynomial quotient is bounded by 2^4 times the other $(1+r)^{-4}$ factor. Its sum is uniformly finite by quadratic volume growth.

B.2 Propagation estimates

We use the bounded-interaction Lieb–Robinson chain argument; see [3, Section 3.1, Theorem 3.1], whose framework allows infinite-dimensional local Hilbert spaces. The proof below records the estimates for our even fermionic algebras and the uniformity in the truncation range ℓ.

Lemma B.1 (Uniform estimate for truncated interactions)**.** *For bounded even ambient operators A, B supported on disjoint cell sets X, Y,*

$$\left\| [\tau_t^{H^{(\ell)}}(A), B] \right\| \le C \, \|A\| \, \|B\| \sum_{x \in X, y \in Y} F_\ell(d(x,y))(e^{v|t|} - 1). \tag{33}$$

The constants are independent of L, ℓ, and of the dimensions of the cell Hilbert spaces. The same estimate holds after restricting the interaction to any collection of its terms.

Proof. For fixed B, let $C_B(X,t)$ be the supremum of $\left\| [\tau_t^{H^{(\ell)}}(A), B] \right\|$ over even $A \in \mathcal{A}_X$ of norm at most one. The interaction-picture commutator equation and Duhamel's formula give, for $t \ge 0$,

$$C_B(X,t) \le C_B(X,0) + 2 \sum_{Z : Z \cap X \neq \varnothing} \left\| \Phi^{(\ell)}(Z) \right\| \int_0^t C_B(Z,s) \, \mathrm{d}s.$$

Terms contained in X can first be included in the local interaction picture; enlarging the resulting boundary sum gives the displayed inequality. Its iteration produces chains $Z_1, \dots, Z_n$ joining X to Y, with ordered time integrals $t^n/n!$. Choose a cell at each intersection. Equation (32) bounds the sum of chain weights by

$$\mathcal{I}_\ell^n C_F^{n-1} \sum_{x \in X, y \in Y} F_\ell(d(x,y)).$$

Since $C_B(Z,0) \le 2\,\|B\|$, summing the series proves (33) with a uniform v. Negative times follow by replacing the interaction by its negative. The proof uses only norms, unitary conjugation, and commutation of disjoint even algebras. It therefore applies to local von Neumann algebra operators and to infinite-dimensional cell spaces. Norm convergence permits finite-sum approximation when needed. □

For a cell set Ω, let $H_\Omega^{(\ell)}$ contain only retained terms whose supports are contained in Ω. Write $X^{(r)} = \{x : d(x,X) \le r\}$. If $r > 2C_{\mathrm{diam}}\ell$, then

$$\begin{aligned} &\left\| \tau_t^{H}(A) - \tau_t^{H^{(\ell)}_{X^{(r)}}}(A) \right\| \\ &\quad \le C_k \, \|A\| \, |t| \, V_L (1+\ell)^{-k} + C \, \|A\| \, |X| V_L \, |t| \, e^{v|t| - r/(2\ell)}. \end{aligned} \tag{34}$$

The first term is the norm Duhamel bound for replacing H by $H^{(\ell)}$. For the second, compare the full and restricted truncated dynamics by Duhamel's formula. Every omitted support Z contains a

cell outside $X^{(r)}$; its diameter is at most $C_{\mathrm{diam}}\ell < r/2$, so $d(X,Z) > r/2$. Lemma B.1, applied to the restricted dynamics and $B = \Phi^{(\ell)}(Z)$, bounds the sum of commutator norms by

$$C\,\|A\|\;|X|e^{-r/(2\ell)}(e^{v|u|}-1)\sum_{Z}|Z|\left\|\Phi^{(\ell)}(Z)\right\| \le C\,\|A\|\;|X|V_L e^{-r/(2\ell)}(e^{v|u|}-1).$$

Integration over $0 \le |u| \le |t|$ gives the second term of (34).

Choose from now on

$$\ell = L^{1/3}, \qquad T = L^{1/3}.$$

For $r = \eta L$, where $\eta > 0$ is fixed, and $|t| \le T$, the right side of (34) is $\|A\|\;O(L^{-\infty})$, uniformly in X. Indeed $|X| \le V_L = O(L^2)$, the first term has an arbitrarily large negative power of ℓ, and the exponential in the second is at most $e^{vL^{1/3}-(\eta/2)L^{2/3}}$. No estimate for a fixed linear light cone of the full interaction is needed.

We next control the separate N_0 evolution. By (6),

$$N_0 = \int_{\mathbb{T}_L} n_R\,\mathrm{d}R, \qquad n_R = \sum_{\sigma} a_\sigma^\dagger(e_R)a_\sigma(e_R),$$

and restricting e_R to $B_r(R)$ gives neutral self-adjoint densities with error Ce^{-cr^2}. Their shell interaction Ψ has Gaussian shell norms. It therefore satisfies (32) with $F_a(r) = e^{-ar}(1+r)^{-4}$ for any fixed $a > 0$. The same chain proof gives the corresponding propagation bound with fixed constants. Denote its restriction to Ω by $(N_0)_\Omega$. In the comparison with $(N_0)_{X^{(r)}}$, omitted terms of diameter less than $r/2$ lie at least $r/2$ from X and are bounded by that propagation estimate. The remaining terms have total norm at most $CV_L(1+r)^{p_0}e^{-cr^2}$, by the Gaussian shells, for a fixed p_0. Duhamel's formula thus gives, uniformly for $|s| \le \pi$,

$$\left\|\tau_s^{N_0}(A) - \tau_s^{(N_0)_{X^{(r)}}}(A)\right\| \le C\,\|A\|\;|X|V_L(1+r)^{p_0}e^{-a_0 r}.$$

In particular this is $\|A\|\;O(L^{-\infty})$ for $r = \eta L$.

Lemma B.2 (Macroscopic locality of the enlarged dynamics)**.** *For fixed $\eta > 0$, $r = \eta L$, and $|t| \le T = L^{1/3}$, a bounded even operator A supported on X has an approximant $\mathcal{L}_{t,X}^{(r)}(A)$ supported on $X^{(2r)}$, such that, in ambient norm,*

$$\left\|\tau_t^{H_b}(A) - \mathcal{L}_{t,X}^{(r)}(A)\right\| \le \|A\|\;O(L^{-\infty}), \qquad \left\|\mathcal{L}_{t,X}^{(r)}(A)\right\| = \|A\|\,.$$

The constants are uniform in t, X, A. The approximants preserve adjoints and neutrality, and are measurable in t. Consequently, if $d(X,Y) \ge cL$ with fixed $c > 0$,

$$\sup_{|t|\le L^{1/3}}\left\|[\tau_t^{H_b}(A), B]\right\| \le \|A\|\,\|B\|\;O(L^{-\infty}) \tag{35}$$

for bounded even A, B supported on X, Y.

Proof. Use the periodic factorization (12), and set

$$\mathcal{L}_{t,X}^{(r)}(A) = \tau_t^{H^{(\ell)}_{X^{(2r)}}}\left(\tau_{s_M(t)}^{(N_0)_{X^{(r)}}}(A)\right).$$

The inner evolution is a local approximation to $\tau_{s_M(t)}^{N_0}(A)$ because $|s_M(t)| \le \pi$. It has support in $X^{(r)}$ and norm $\|A\|$. Apply (34) to this operator with a further buffer r to approximate the outer H

evolution. The triangle inequality proves the stated error. Both restricted Hamiltonians are neutral and self-adjoint. Their unitary conjugations give the norm and adjoint assertions. The chosen representative $s_M(t)$ is measurable, so the approximant is measurable; its jumps do not affect any estimate or integral. The restricted evolutions need neither preserve the LLL nor commute with each other. Finally choose $2\eta < c$; the local approximant commutes with B, proving (35). □

B.3 Boundary currents and unbounded charges

The charge $Q = Q_\Gamma$ is unbounded on ambient Fock space. A small norm error in a local approximation cannot be commuted with Q directly. Finite LLL rank provides a bounded substitute for this commutator.

Lemma B.3 (Finite charge bandwidth and a bounded current map)**.** *Let $\alpha_\theta(A) = e^{\mathrm{i}\theta Q} A e^{-\mathrm{i}\theta Q}$. For every operator A in the finite LLL CAR algebra, extended canonically to ambient Fock space,*

$$\alpha_\theta(A) = \sum_{|n|\le M} e^{\mathrm{i}n\theta} A_n.$$

There is a bounded linear map $\mathcal{D}_M$ on ambient bounded operators such that

$$\|\mathcal{D}_M(A)\| \le CM \|A\|, \qquad \mathcal{D}_M(A) = \mathrm{i}[A, Q] \quad \text{for LLL operators } A.$$

The latter commutator has a bounded extension. The map preserves physical support, neutrality, and self-adjointness.

Proof. On the finite-particle core,

$$[Q, a_\uparrow(f)] = -a_\uparrow(1_\Gamma f), \qquad [Q, a_\uparrow^\dagger(f)] = a_\uparrow^\dagger(1_\Gamma f).$$

Equivalently,

$$\alpha_\theta(a_\uparrow^\dagger(f)) = a_\uparrow^\dagger(1_{\Gamma^c} f) + e^{\mathrm{i}\theta} a_\uparrow^\dagger(1_\Gamma f),$$

with the conjugate phase for annihilators and no phase for spin-down fields. Choose any basis of the M spin-up LLL orbitals. Normal-ordered monomials span the finite LLL CAR algebra, and each contains at most M spin-up creators and at most M spin-up annihilators. Expanding the displayed identity in each monomial gives frequencies in $[-M, M]$. No estimate on the number or coefficients of these monomials is used.

Choose a real even $\rho \in C_c^\infty(\mathbb{R})$, equal to one on $[-1, 1]$ and supported in $[-2, 2]$, and set

$$\kappa_M(\theta) = \sum_{n\in\mathbb{Z}} \rho(n/M) e^{-\mathrm{i}n\theta}.$$

To record its uniform bounds, put $\check{\rho}(u) = (2\pi)^{-1} \int \rho(\xi) e^{-\mathrm{i}u\xi} \, \mathrm{d}\xi$. Periodization, or comparison of Fourier coefficients, gives

$$\kappa_M(\theta) = 2\pi M \sum_{j\in\mathbb{Z}} \check{\rho}\big(M(\theta + 2\pi j)\big).$$

The Schwartz decay of $\check{\rho}$ and its derivative therefore implies

$$\frac{1}{2\pi}\int_0^{2\pi} |\kappa_M(\theta)| \, \mathrm{d}\theta \le C, \qquad \frac{1}{2\pi}\int_0^{2\pi} |\kappa_M'(\theta)| \, \mathrm{d}\theta \le CM.$$

Define

$$\mathcal{D}_M(A) = \frac{1}{2\pi}\int_0^{2\pi} \kappa_M'(\theta) \alpha_\theta(A) \, \mathrm{d}\theta.$$

For arbitrary bounded operators this is a weak operator integral: all matrix elements of the integrand are continuous, and the preceding bound defines a bounded operator of norm at most $CM\,\|A\|$. Norm continuity of the gauge action on arbitrary local approximants is not required. On each of the finitely many Fourier modes of an LLL operator, integration by parts gives

$$\mathcal{D}_M(A) = \sum_{|n|\le M} (-\mathrm{i}n)A_n = -\left.\frac{\mathrm{d}}{\mathrm{d}\theta}\alpha_\theta(A)\right|_{\theta=0} = \mathrm{i}[A,Q].$$

The last identity holds on the finite-particle core; the middle expression supplies its bounded extension. Regional gauge rotations preserve local von Neumann algebras and neutrality, and these properties pass to the weak integral. Since κ'_M is real, self-adjointness is preserved as well. □

Lemma B.4 (Boundary currents)**.** *There are bounded self-adjoint neutral operators $J_\pm$ with*

$$\mathrm{i}[H_b, Q] = J_- + J_+, \qquad \|J_\pm\| \le CML_y.$$

For every $r \ge 1$ they have self-adjoint neutral approximants $J_\pm^{(r)}$ supported in the strips of transverse radius r about $\partial_\pm$, satisfying, for every k,

$$\left\|J_\pm - J_\pm^{(r)}\right\| \le C_k ML_y(1+r)^{-k}. \tag{36}$$

All these are ambient norm bounds.

Proof. Write $H = H_{\mathrm{LLL}}$ and define

$$j_{H,R} = \mathcal{D}_M(h_R) = \mathrm{i}[h_R, Q], \qquad j_{H,R}^{(r)} = \mathcal{D}_M(h_R^{(r)}).$$

The preceding lemma and Definition 2.3 give

$$\|j_{H,R}\| \le CM, \qquad \left\|j_{H,R} - j_{H,R}^{(r)}\right\| \le C_k M(1+r)^{-k}, \qquad \operatorname{supp} j_{H,R}^{(r)} \subset B_r(R).$$

If this ball lies wholly inside Γ, its local neutral operator $h_R^{(r)}$ is fixed by α_θ; the same holds if the ball lies wholly outside Γ. In either case $\mathcal{D}_M(h_R^{(r)}) = 0$. Choose $r = \frac{1}{2}d(R,\partial\Gamma)$ when that distance is at least two, and use the uniform bound otherwise. This proves

$$\|j_{H,R}\| \le C_k M\big(1 + d(R,\partial\Gamma)\big)^{-k}. \tag{37}$$

Let $\Omega_\pm$ partition the centers according to the nearer boundary component, assigning the measure-zero ties arbitrarily, and set

$$J_{H,\pm} = \int_{\Omega_\pm} j_{H,R}\,\mathrm{d}R.$$

Integration of (37) in the transverse direction gives $\|J_{H,\pm}\| \le CML_y$, and $J_{H,-} + J_{H,+} = \mathcal{D}_M(H) = \mathrm{i}[H,Q]$. For $2 \le r < L_x/2$, use the explicit local approximation

$$J_{H,\pm}^{(r)} = \int_{\Omega_\pm \cap \{d(R,\partial_\pm)\le r/2\}} j_{H,R}^{(r/2)}\,\mathrm{d}R.$$

Every integrand is supported in the indicated strip of radius r. The omitted centers contribute at most $C_k ML_y(1+r)^{1-k}$ by (37); the retained centers have area at most $CL_y r$, so their approximation errors satisfy the same bound. Increasing k proves the analogue of (36) for $J_{H,\pm}$. For $1 \le r < 2$ one

may use zero; when $r \geq L_x/2$, the strip covers the torus and one uses the exact current. The same estimates apply. If the given local approximants are only weakly measurable, all center integrals are taken weakly. Uniform norm bounds, scalar Fubini, and the weak closure of the local algebras justify the formulas and preserve support and neutrality.

For N_0, the resolution of Π in (6) gives a sharper one-body construction. Its current density is

$$j_{0,R} = \mathrm{i}(a_\uparrow^\dagger(e_R)a_\uparrow(1_\Gamma e_R) - a_\uparrow^\dagger(1_\Gamma e_R)a_\uparrow(e_R)).$$

Inside Γ, replace $1_\Gamma e_R$ by $e_R - 1_{\Gamma^c} e_R$; the two terms with e_R cancel. Outside Γ, $1_\Gamma e_R$ is already a tail. Thus (6) yields $\|j_{0,R}\| \leq Ce^{-cd(R,\partial\Gamma)^2}$. Cutting every test vector off at radius r gives a local approximation with Gaussian error. Integration over $\Omega_\pm$ and the same center cutoff as above yield

$$\begin{aligned} \mathrm{i}[N_0, Q] &= J_{0,-} + J_{0,+}, \qquad \|J_{0,\pm}\| \leq CL_y, \\ \left\|J_{0,\pm} - J_{0,\pm}^{(r)}\right\| &\leq CL_y(1+r)^c e^{-cr^2}, \end{aligned}$$

with possibly different positive constants in the last estimate. Finally set

$$J_\pm = J_{H,\pm} - \mu_M J_{0,\pm}, \qquad J_\pm^{(r)} = J_{H,\pm}^{(r)} - \mu_M J_{0,\pm}^{(r)}.$$

Since $\mu_M = O(M)$, these have the claimed norm and rapid locality bounds. The commutator identities hold on the finite-particle core and, in particular, on every fixed-number sector, where the regional charge is bounded. □

C Filters, dressing, and clustering in a sector

C.1 The inverse-energy filter

Let $\Delta = \gamma > w$ be the gap bound retained by Lemma 2.5. In the selected number sector, H_b has band energies in $[E_1 - \mu_M N, E_d - \mu_M N]$, with the same width and gap. Write $\tau_t = \tau_t^{H_b}$. Choose a real smooth even function χ equal to one near zero and zero for $|\omega| \geq \Delta$. With Fourier convention $\widehat{W}(\omega) = \int W(t)e^{\mathrm{i}t\omega}\,\mathrm{d}t$, choose

$$\widehat{W}(\omega) = \frac{1 - \chi(\omega)}{\mathrm{i}\omega}.$$

Its inverse Fourier transform is real, bounded, and integrable, with

$$\int (1 + |t|)^m \, |W(t)| \, \mathrm{d}t < \infty \quad \text{for every } m. \tag{38}$$

For an explicit construction, let

$$\check{\chi}(t) = \frac{1}{2\pi} \int \chi(\omega) e^{-\mathrm{i}t\omega} \, \mathrm{d}\omega.$$

This is a real even Schwartz function with $\int \check{\chi}(t)\,\mathrm{d}t = 1$. For $t \neq 0$, set

$$W(t) = -\operatorname{sgn}(t) \int_{|t|}^\infty \check{\chi}(s)\,\mathrm{d}s, \qquad W(0) = 0. \tag{39}$$

It is real, odd, and bounded, with one-sided limits $W(0^\pm) = \mp 1/2$. Its tail integral decays faster than every power, proving (38). Distributionally, $W' = \check{\chi} - \delta_0$, so $-\mathrm{i}\omega\widehat{W}(\omega) = \chi(\omega) - 1$. This proves the asserted Fourier formula, with its continuous extension at $\omega = 0$; oddness gives $\widehat{W}(0) = 0$.

Define bounded ambient operators

$$K_{\pm}^{\infty} = \int_{\mathbb{R}} W(t)\tau_t(J_{\pm})\,\mathrm{d}t, \qquad K^{\infty} = K_{-}^{\infty} + K_{+}^{\infty}.$$

The integrals converge in norm. On the fixed-number sector, matrix elements between the low-energy band and its orthogonal complement have energy difference of magnitude at least Δ. For such a difference ω, $\widehat{W}(\omega)\mathrm{i}\omega = 1$. The two off-diagonal blocks relative to P therefore give

$$[K^{\infty}, P] = [Q, P]. \tag{40}$$

This can equivalently be checked using spectral measures; no finite-dimensionality of the enlarged sector is required. Only the two off-diagonal blocks relative to P enter this identity. The inverse-energy filter itself requires only $\Delta > 0$, with no restriction on the band width.

To localize the dressings, fix $r = \eta L$, with $\eta > 0$ small enough that the two boundary strips remain separated. Approximate $J_{\pm}$ in the strip of radius $r/4$, using (36). On $|t| \le T = L^{1/3}$, apply the composed local approximation of Lemma B.2, with nested strips and buffers proportional to r. Denote the resulting operator by $G_{\pm,r}(t)$: it is supported in the strip of radius r, is self-adjoint and neutral, and satisfies

$$\sup_{|t|\le T} \|\tau_t(J_{\pm}) - G_{\pm,r}(t)\| = O(L^{-\infty}), \qquad \|G_{\pm,r}(t)\| \le CML_y.$$

For example, first use restricted N_0 dynamics in the strip of radius $r/2$, then restricted range-$L^{1/3}$ dynamics of H in the strip of radius $3r/4$; cell padding fits inside radius r for large L. Neither restricted generator needs to preserve the LLL or to commute with the other. For each finite torus the restricted generators are bounded, so these operators are norm continuous between the finitely many jumps of $s_M(t)$ on $[-T, T]$. Their time integrals therefore converge in norm. Set

$$K_{\pm}^{(r)} = \int_{-T}^{T} W(t)G_{\pm,r}(t)\,\mathrm{d}t.$$

The local-approximation error is $O(L^{-\infty})$ in norm. For the omitted times, (38) gives, for every m,

$$\left\| \int_{|t|>T} W(t)\tau_t(J_{\pm})\,\mathrm{d}t \right\| \le CML_y \int_{|t|>T} |W(t)|\,\mathrm{d}t \le C_m L^3 T^{-m}.$$

Consequently, for each fixed admissible η,

$$\operatorname{supp} K_{\pm}^{(r)} \subset \{d(\cdot, \partial_{\pm}) < r\}, \qquad \left\|K_{\pm}^{(r)}\right\| = O(L^3), \qquad \left\|K_{\pm}^{\infty} - K_{\pm}^{(r)}\right\| = O(L^{-\infty}). \tag{41}$$

These are ambient norm estimates, and the approximants are self-adjoint and neutral. Take $r = L_x/64$; bounded aspect ratio makes the estimates uniform for this choice. Then (40) implies the required commutation estimate for $K_{\pm} = K_{\pm}^{(r)}$. A magnetic translation moves their supports by $\delta = O(L^{-1})$, leaving ample margin inside $B_{\pm}$; its conjugates supply $K_{\pm}^{U}$.

C.2 Clustering across the low-energy projection

We give a filter proof that uses a gap only in the selected sector. Choose $\vartheta \in C^{\infty}(\mathbb{R})$ with

$$\vartheta(\omega) = 1 \quad (\omega \le -\Delta), \qquad \vartheta(\omega) = 0 \quad (\omega \ge -(w + \Delta)/2).$$

For a bounded neutral local A, write $A_\vartheta = \vartheta(\mathrm{ad}_{H_b})A$. This is unambiguous in finite volume: H_b acts on the finite-dimensional LLL Fock factor and as the identity on its complement, so it has finitely many distinct eigenvalues. If $H_b = \sum_\lambda \lambda E_\lambda$, set

$$A_\vartheta = \sum_{\lambda,\lambda'} \vartheta(\lambda - \lambda') E_\lambda A E_{\lambda'}.$$

For matrix elements within $\operatorname{Ran} P$, the energy difference belongs to $[-w, w]$, where $\vartheta = 0$ because $w < \Delta$ implies $-w > -(w + \Delta)/2$. For a matrix element with output in $\operatorname{Ran} P$ and input in its complement, the energy difference is at most $-\Delta$, where $\vartheta = 1$. For the reverse matrix element it is at least Δ, where $\vartheta = 0$. Neutrality restricts these statements to the selected number sector. Thus, even with distinct energies inside the band, the exact identities are

$$PA_\vartheta = PA(1 - P), \qquad A_\vartheta P = 0.$$

Consequently, for another bounded neutral local B,

$$PA(1 - P)BP = P[A_\vartheta, B]P. \tag{42}$$

The identities are sector identities; ϑ need not remove transitions in other charge sectors.

We estimate the distributional Fourier representation directly. Since ϑ' is smooth and compactly supported, the inverse transform of ϑ is a multiple of $\delta(t)$ plus $\beta(t)/t$ in principal value, with β Schwartz. For disjoint supports $[A, B] = 0$, so the delta term vanishes in $[A_\vartheta, B]$. The integral at zero is norm convergent: H_b is bounded and the fundamental theorem of calculus gives

$$\|[\tau_t(A), B]\| \le 4|t| \, \|H_b\| \, \|A\| \, \|B\| \le CL^4|t| \, \|A\| \, \|B\| \, . \tag{43}$$

This estimate uses H_b directly and does not differentiate $s_M(t)$.

Suppose the physical supports are separated by at least cL, where $c > 0$ is fixed. For a desired error L^{-k}, choose $a > k + 4$, set $\delta_L = L^{-a}$, and split the filter integral into three ranges. For $|t| \le \delta_L$, (43) cancels its $1/|t|$ singularity and bounds the contribution by $C \, \|A\| \, \|B\| \, L^4 \delta_L$. For $\delta_L < |t| \le T = L^{1/3}$, Lemma B.2 bounds the commutator by $C_m \, \|A\| \, \|B\| \, L^{-m}$ for every m. The integral of $1/|t|$ over this range costs only $O((a + 1) \log L)$, so choosing $m > k + 1$ suffices. For $|t| > T$, the trivial commutator bound $2 \, \|A\| \, \|B\|$ and the Schwartz tail of β give $C_j \, \|A\| \, \|B\| \, T^{-j}$ for every j. Choosing $j > 3k$ and using (42) proves

$$\|PA(1 - P)BP\| \le C_{k,c} \, \|A\| \, \|B\| \, L^{-k} \qquad \text{for every } k. \tag{44}$$

All support and volume factors have been absorbed using the uniform $O(L^2)$ cell count in Lemma B.2. The estimate also holds for bounded local von Neumann algebra operators, including regional-charge exponentials: the propagation proof uses only bounded local operators and commuting disjoint even algebras. For S_-, S_+ this is exactly (4).

C.3 Bookkeeping for sector-bounded charges

The only operators in the index proof that lack a bounded ambient norm are regional charges, the corresponding translation transports, and their sums with bounded dressings. Their restrictions to $\mathcal{H}_n$ have polynomial norms. All errors involving those operators are estimated after restriction, for example

$$\|[Q_X, E]|_{\mathcal{H}_n}\| \le 2n_\uparrow \, \|E|_{\mathcal{H}_n}\|$$

for neutral E. Their exponentials, on the other hand, are ambient local unitaries of norm one; these are the operators to which the clustering and separated-factor arguments are applied. Bounded perturbations of a local charge are self-adjoint on its domain and have local unitary exponentials. Thus no step applies a Lieb–Robinson estimate with an unbounded ambient observable, and no step regards a global particle-number compression as a local operation.

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